Olena Teslia

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Education

Columbia University, School of Engineering & Applied Sciences, NY Aug. 2024 – May 2028

- Bachelor of Science in Electrical Engineering, minors in Applied Mathematics, Computer Science; GPA: 3.81/4.00.
- Relevant Coursework: Quantum Computing Lab, Fundamentals of Computer Systems, Signals and Systems, Probability Theory, Electronic Circuits, Electromagnetics, Solid State Devices, Computing with Brain Circuits.
- Extracurricular: NeurotechX, FormulaSAE, Pi Mu Epsilon, Hackhouse, EcoReps, AI Safety, Mentor Initiative.
- Work: Laboratory Assistant, Mentor Chair, Student Worker.

Experience

Columbia Fusion Research Center, *Research Assistant*, NY Aug. 2025 – Present

- Designing, wiring, and testing power, control, and safety systems 20+ hours/week for the Tokamak for Education.
- Developed an interlocked winding system and a Hall probe magnetic field sensor for the Stellarator Experiment.
- Collaborated in assembly of the vacuum chamber setup, CAD modeling, and machining.

Brave1, *Components Intern*, Ukraine Jun. 2026 – Sep. 2026

- Interviewed 10+ companies and distributors, conducted research on the supply chain, collecting 40+ responses through the questionary and creating the database, organized a webinar for 30+ participants.

Program on Artificial Intelligence and Reasoning, *Junior Counselor*, UK Aug. 2023, 2025

- Assisted a 10-day workshop with 60+ hours studying mathematics, cognition, and AI/ML for 25 participants from 1200+ applicants; led a lecture on the brain, organized 7+ activities, and worked in a 15-people staff team.

Projects

Drone Journalist | *Raspberry PI, Linux, Bash, Python, AI* Oct. 2025

- Worked on a vision for the journalist drone, assisted with voice processing debugging, and an assembly.

Memory Game | *Arduino, C++, Onshape, machining* Oct. 2024 – Dec. 2024

- Developed an Arduino-based memory game with button-screen-speaker IO interface.
- Used 3D-printing and laser-cutting for manufacturing and Visual Studio and ArdinoIDE.

Guitar Learning App | *Python, Kotlin, MySQL, SQLight, Node.js* Jun. 2023 – May 2024

- Developed a full-stack Android application serving a REST API with Kotlin as the frontend using Android Studio.
- Processed sound and images using FFTs and OpenCV attempting at Optical Music Recognition implementation.
- Created multi-table local (ROOM) and global (MySQL) databases, and an IO user interaction system.

Leadership

Ukraine Global Scholars, *Mentor*, Ukraine Jul. 2021 – Present

- Screened 90+ applicants for UGS (a non-profit that supports talented underprivileged students) over 3 stages.
- Co-led 10-day boot camp for 50+ students, organizing daily lectures, fun, and well-being activities.
- Advised 15+ applicants to colleges, editing their essays, CVs, forms, brainstorming, and doing mock interviews.

Engineering Without Borders, *Fundraising Co-Lead*, New York, NY Sep. 2024 – Jan. 2026

- Reached out to companies, co-organized fundraisers, applied for grants, and conducted weekly meetings.

International Mathematics Olympiad, *Volunteer*, Bath, UK Jul. 2024

- Guided a team of 6 participants throughout the 10-day event; organized excursions for 100+ people, 10+ hours; welcomed 80+ teams with total of 500+ members, 20+ hours; helped with logistics.

Skills

Software: SolidWorks, Fusion360, OnShape, Altium, LTspice, Microsoft Suit, Visual Studio, Git, Linux, Windows.
Programming: Python, Java, SQL, Kotlin, MATLAB, JavaScript, LaTeX, C++, Haskell, Bash.
Machining: 3D printing (PLA, ABS), waterjet, laser cutter, vertical bandsaw, electronics, woodshop.
Art: 9+ years in Visual Arts (watercolor, pastel, graphics), 6+ years in Poetry (writing, translation).